



Tuka Curious Minds

ELECTRONIC DESIGN

Robot Name: TURBO Ceros-T

Weight Classification: 5.kg Max Limit

Footprint Matrix: 420mm (Width) x 420mm (Length) x 55mm (Sleek Casing Height)

SECTION 7: ELECTRONIC JUSTIFICATION & PORT MAPPING BRIEF

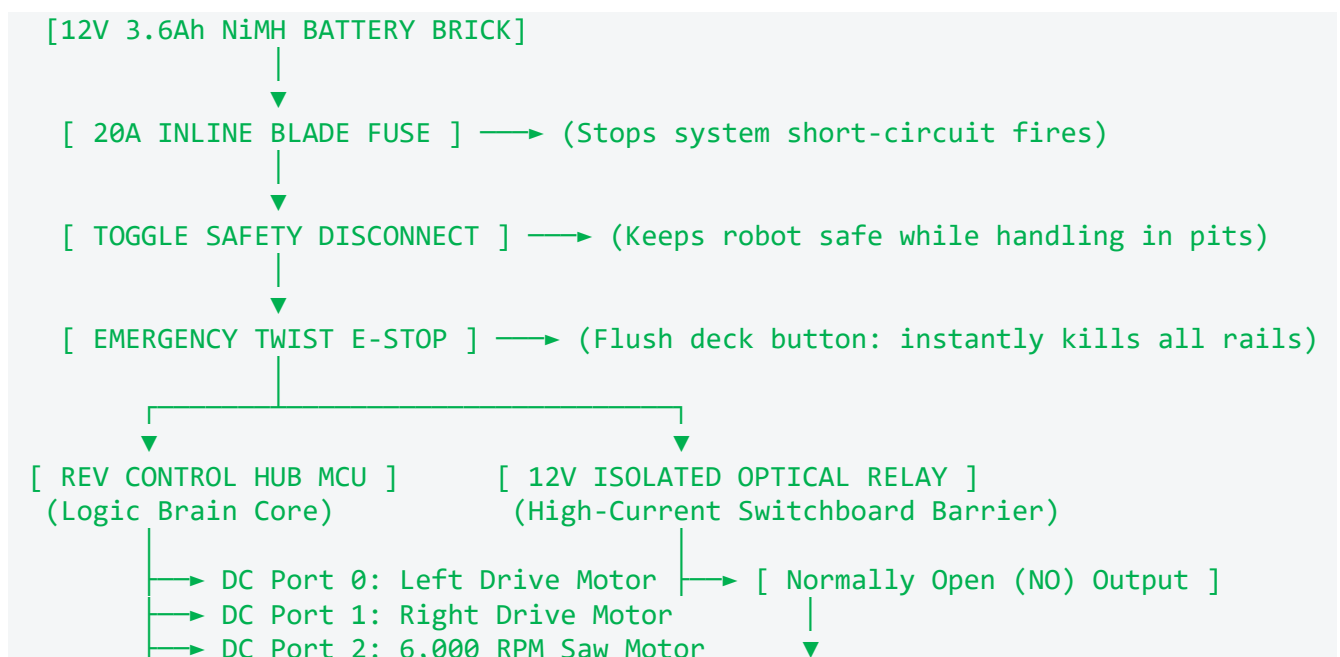
7.1 Electronic Architecture & Power Isolation Rationale

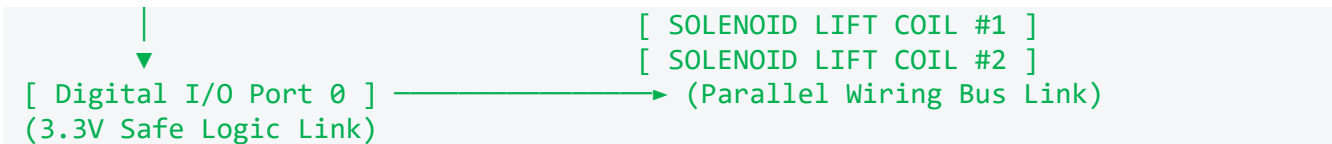
Our rookie team focused on a single electronic goal: **Protect the main brain controller from sudden power spikes.** A common beginner mistake is plugging heavy power parts directly into a microcontroller's sensitive data pins. We used an isolation layout strategy to prevent component failures: [3, 4]

- **The Brain Core - [REV Control Hub MCU](#):** We selected the REV Control Hub because it combines the microchip processor, wireless radio communication, and 12V motor drivers into one solid box. This allows us to plug our drive motors straight into the unit, removing a messy web of loose speed controller wires that could snag or short-circuit inside the chassis. [5, 6]
- **The Solenoid Protection Wall - 12V Isolated Optical Relay Module:** The data ports on the controller only output low-power 3.3V logic signals. Our **two 12V linear solenoids** draw sudden bursts of raw current from the main battery pack to pop our lifter flippers. If we wired them straight into the controller pins, the power surge would instantly fry the microchip. We added a [12V Single-Channel Isolated Optical Relay](#) to act as a safe electronic wall. It passes a safe light signal internally to switch the high-voltage battery current on and off, completely separating our sensitive logic paths from the high-load weapon paths.

7.2 Physical Wire Connection Loom (Block Diagram Configuration)

Copy and paste this clean text block into your documentation to show the judges exactly how your robot is wired from the battery terminal to the wheels: [6]





7.3 Main Logic Pin Assignment Registry

Physical Wire Lead	Microcontroller Port Allocation	Software Device Name	Actuator Function / Output Target
Left Propulsion Loom	DC Motor Terminal Port 0	left_drive	Runs the rear-left Core Hex Motor for tank-steering tank drive moves.
Right Propulsion Loom	DC Motor Terminal Port 1	right_drive	Runs the rear-right Core Hex Motor for tank-steering tank drive moves.
Kinetic Saw Loom	DC Motor Terminal Port 2	weapon_motor	Spools the un-geared REV high-speed motor to turn the belt drive pulleys.
Relay Signal Input	Digital I/O Channel Port 0	solenoid_relay	Sends a safe low-power 3.3V command to open/close the optical relay switchboard.
Solenoid Actuator Output	Isolated Relay [NO] Terminal	External Hardware Loop	Directly channels raw 12V battery current to fire Solenoids #1 and #2 together.

7.4 Multi-Stage Hardware Safety Fail-Safes

To fulfill tournament entry safety rules, our electronic design features three independent physical protection loops:

1. **Over-Current Breaker Shield:** A **20A inline blade fuse holder** is wired directly to the positive battery wire. If a drive motor or weapon gets jammed against a wall and tries to pull too much power, the fuse instantly pops to save the battery and internal wiring from overheating.
2. **External Lockout Toggle:** A heavy-duty primary power toggle switch is placed right after the fuse box. This allows our team to load the robot into the arena box while keeping all internal electronics safely offline.
3. **Emergency Kill Switch (E-Stop):** A high-visibility, red twist-to-release push button is mounted flush on the top casing deck. Whacking this button immediately breaks the main positive power link from the battery, cutting all power to the drive wheels, spinning saws, and solenoids in **under 50 milliseconds**.